

Week 1

1. What is Power BI?

Power BI is a business intelligence tool used to:

- Import data
- Clean and transform data
- Build data models
- Create interactive reports & dashboards
- Share insights

Core Purpose

Convert **raw data** → **meaningful visuals** → **business decisions**

Import Data:

Open Power BI Desktop → Home → Get Data → choose source (Excel/CSV/SQL) → select file/table → choose *Load*

-Load only if data is transformed (cleaned)

Clean:

-Cleaning data means removing unnecessary columns, renaming headers, changing data types (text, whole number, or decimal), removing blanks, and filtering rows.

- Then Home → Close & Apply.

Week 2: Data Cleaning Using Python

-Purpose: data manipulation and numerical operations

- Import pandas, numpy
- Load CSV dataset
- Check shape and basic info

- Remove duplicate Ticket_ID records
- Standardize text columns (lowercase, strip spaces)
- Fix wrong category values (typos)
- Convert numeric columns using `pd.to_numeric()`
- Replace invalid numeric values with NaN
- Handle missing values using median (group-wise where needed)
- Apply business rules ($\text{Resolved} \leq \text{Assigned}$, $\text{Reopened} \leq \text{Resolved}$)
- Create derived columns (SLA_Met, Resolution_Efficiency, First_Time_Resolution)
- Validate cleaned data
- Save cleaned dataset to CSV

Week 3 – Exploratory Data Analysis (EDA)

- Identify dataset size using total titles, movies, and TV shows
- Calculate distribution of content type (Movies vs TV Shows)
- Analyze trend of content releases by year
- Find top contributing countries by number of titles
- Identify most common genres
- Analyze rating distribution for Movies and TV Shows
- Compare content growth across time
- Detect dominance of Movies over TV Shows
- Observe content surge after 2015
- Identify countries producing majority of content
- Detect genre concentration (Drama, Comedy, Documentaries)
- Compare rating patterns between Movies and TV Shows
- Summarize key patterns and anomalies
- Document business insights from visuals

Week 4 – Live Data Import in Power BI , APIs

- Understand static data vs live data
- Understand what an API is and why it is used
- Generate API key / API token from provider
- Store API key securely
- Open Power BI → Get Data → Web
- Choose authentication type (Web API / Basic / Anonymous)

- Pass API key via header or URL parameter
- Load JSON response
- Expand lists and records in Power Query
- Rename columns
- Change data types
- Apply Close & Apply
- Configure scheduled refresh in Power BI Service

Zendesk

- Customer support ticketing platform
- Converts emails, chats, calls, forms into tickets
- Tracks ticket status (New → Open → Pending → Solved → Closed)
- Stores users, agents, and organizations
- Measures support metrics (volume, resolution time, SLA, CSAT)
- Used for monitoring support performance and customer experience.

Week 5 – KPIs & Data Modeling

KPIs (Key Performance Indicators):

- KPI means **Key Performance Indicator**.
- It is a measurable value that shows how well a business is performing.
- KPIs help in checking whether goals are being achieved or not.
- A good KPI must be clear, measurable, and related to business objective.
- Each KPI usually has:
 - Actual value
 - Target value
 - Status (Good/Bad)
- KPIs are created using Card visual or KPI visual in Power BI.
- Measures are used to calculate KPIs.
- KPIs are used to:
 - Track performance
 - Identify problems
 - Support decision making

- **KPI :**
- Total Sales
- Profit Margin
- Average Resolution Time
- SLA Compliance %
- Customer Satisfaction Score
- **-Data Modeling in Power BI**
- Data modeling means organizing tables so they work correctly together.
- It helps in getting accurate totals and correct KPIs.
- A good data model avoids duplicate counting and wrong results.
- **Fact Table**
- Contains numerical values.
- Has many rows.
- Example: Sales, Orders, Tickets.
- **Dimension Table**
- Contains descriptive information.
- Has fewer rows.
- Example: Customers, Products, Dates, Agents.
- Primary Key is a unique column in dimension table.
- Foreign Key is the same column in fact table used for connection.
- Data type of both keys must be same.
- Most common relationship type is One-to-Many.
- Dimension table is on “One” side.
- Fact table is on “Many” side.
- Use single direction filter in relationships.
- Use both direction only if required.
- Best structure is Star Schema.
- One fact table in center.
- Use Measures for totals and KPIs.

Week 6- DAX (Data Analysis Expressions)

- DAX stands for **Data Analysis Expressions**.
- It is a formula language used in Power BI to perform calculations on data.
- DAX is mainly used to create **Measures, Calculated Columns, and Tables**.
- Used to calculate totals, averages, and percentages
- Used to build KPIs

- Used to apply business logic
- Helps convert raw data into meaningful insights

Measures

- Calculated at report level
- Change based on filters and slicers
- Used for totals and KPIs

Calculated Columns

- Calculated row by row
- Stored inside table
- Used for categorization or flags

Filter Context (Very Important)

- Filters decide **which data** is used in calculation
- Slicers, visuals, and CALCULATE change filter context
- Wrong filter context = wrong result

CALCULATE Function

- Most powerful DAX function
- Changes filter context

Week 7-DATA VISUALIZATION

-DATA VISUALIZATION-Used for

- Graphical representation of data
- Converts numbers into visual form
- Helps find patterns, trends, and comparisons and also for Faster understanding, Easy comparison, Better decision making

Bar Chart

- Compares categories
- X-axis → Categories
- Y-axis → Values

Use for: Product sales, department comparison

Do not use for: Time trends

Line Chart

- Shows change over time
- X-axis → Time
- Y-axis → Value

Use for: Monthly sales, yearly growth

Do not use for: Category comparison

Pie Chart

- Shows part of whole
- Total = 100%
- Max 5 slices

Use for: Market share, budget split

Box Plot

- Box plot shows **distribution, spread, median, and outliers**
- Based on five-number summary

Minimum, Q1, Median, Q3, Maximum

Box → Q1 to Q3

Line → Median

Whiskers → Min & Max

Dots → Outliers

$$\text{IQR} = Q3 - Q1$$

Outlier limits:

$$\text{Lower} = Q1 - 1.5 \times \text{IQR}$$

$$\text{Upper} = Q3 + 1.5 \times \text{IQR}$$

Week 8-Dashboards

Dashboard 1: Operational Overview

Purpose: This tab focuses on the "What"—ticket volumes, adherence to deadlines (SLAs), and the nature of the issues coming in.

1. Top KPI Cards

Total Tickets Assigned (384): The total workload that came into the queue.

Total Tickets Resolved (344): The number of tickets closed.

Insight: The team has cleared ~90% of the assigned volume, which is a strong closure rate.

Open Tickets (40): What is currently pending in the pipeline.

SLA Compliance % (56%): The percentage of tickets resolved within the agreed-upon time limit.

Insight: This is a critical red flag. Ideally, this should be 90%+. A 56% compliance means nearly half of all tickets are breaching deadlines.

Average Resolution Time (5.81 Hours): It takes the team nearly 6 hours on average to fix an issue.

1. Customer Satisfaction by Issue and Priority (Top Left - Combo Chart)

Visual: Blue bars with a line chart overlay.

What it Measures:

Bars (Left Axis): The number of tickets (Volume) per Issue Type (Software, Database, Hardware, etc.).

Line (Right Axis): The Average Customer Satisfaction (CSAT) score for those issues.

Key Insight: "Software" has the highest volume (tallest bar), which usually puts the most strain on the team. If the line drops lower on the 'Software' bar, it means high volume is negatively affecting customer happiness.

2. Ticket Distribution by Priority (Center - Column Chart)

Visual: Vertical blue bars sorted by height.

What it Measures: Breaks down all tickets by urgency: High, Medium, Low, Critical.

Key Insight: The tallest bars are **High** and **Medium**.

Talking Point: "Our team is in 'fire-fighting' mode. The majority of our work is tagged as High Priority, which makes it very difficult to manage our SLA compliance because everything is treated as urgent."

3. Count of Ticket_ID by Issue_Type (Bottom Left - Donut Chart)

Visual: A colorful ring chart.

What it Measures: The percentage of total tickets belonging to each category.

Key Insight:

Software (24%) and **Database (20%)** make up nearly half the workload.

Talking Point: "Since Software and Database issues are nearly 50% of our intake, any training we do should focus on these two areas to get the biggest ROI."

4. SLA Compliance by Issue (Bottom Center - Bar Chart)

Visual: Horizontal bars.

What it Measures: How often we meet the deadline (SLA) for each specific issue type.

Key Insight:

Hardware appears to have the shortest bar (lowest compliance).

Talking Point: "We are struggling most with Hardware tickets. This might be due to external factors like waiting for parts to ship, rather than agent performance."

5. Detailed Ticket Grid (Bottom Right - Table)

Visual: A spreadsheet-like list.

What it Measures: Granular data for every single ticket (ID, Agent, Priority, Status).

Key Insight: This is for "Drill Down." If a manager asks, "Which specific tickets failed SLA?", you scroll through this list to find the rows where "SLA_Met" says "No."

Dashboard 2: Agent Performance and Quality Analysis

Purpose: This tab focuses on the "Who"—comparing individual team members to identify top performers and those needing coaching.

1. Top KPI Cards

Resolution Efficiency % (89.6%): Overall, the team resolves almost 90% of what they touch.

Avg Resolution Time (5.81 hrs): Consistent with the first dashboard.

SLA Compliance (56.0%): Consistent with the first dashboard.

Backlog Tickets (40): The number of unresolved tickets currently sitting with agents.
(Note: There is a typo in the dashboard "Backlog Tickes"—you might want to mentally correct this while speaking).

1. Resolution Efficiency % by Agent_ID (Top Left - Bar Chart)

Visual: Horizontal blue bars ranking agents.

What it Measures: The percentage of assigned tickets an agent successfully resolved without error or delay.

Key Insight:

Top: Agents **A102** and **A105** are tied at **95%**.

Bottom: Agent **A103** is at **74%**.

Talking Point: "There is a 20% performance gap between our best agent (A102) and our struggling agent (A103)."

2. Average Resolution Time by Agent_ID (Top Center - Bar Chart)

Visual: Horizontal blue bars showing time.

What it Measures: The average hours it takes each agent to close a ticket.

Key Insight:

Agent A103 has the longest bar (slowest).

Agent A101/A102 have shorter bars (fastest).

Talking Point: "Not only is A103 less efficient, but they also take significantly longer to resolve issues, which drags down our team average."

3. Agent vs Issue Type Workload (Bottom Left - Matrix Table)

Visual: A grid of numbers with blue heat-map shading.

What it Measures: Who is doing what. (e.g., How many Database tickets did A101 handle vs A104?).

Key Insight:

Look at the "Total" column on the far right. Everyone has roughly **5 tickets**.

Talking Point: "This chart proves that the workload is distributed fairly. Agent A103 isn't performing poorly because they have *more* work; they have the same amount as everyone else."

4. Ticket Reopen Rate by Agent (Bottom Center - Stacked Column Chart)

Visual: Vertical bars with two colors (Light Blue = Not Reopened, Dark Blue = Reopened).

What it Measures: Quality Control. "Reopened" means the agent said it was fixed, but the customer said it wasn't.

Key Insight:

Agent A103 has a very large **Dark Blue** section.

Other agents have almost no dark blue.

Talking Point: "This is the smoking gun. Agent A103 is closing tickets prematurely, causing customers to reopen them. This 'double handling' is why their efficiency is low and why our overall SLA is suffering."

5. Slicers (Right Side - Interactive Buttons)

Visual: Buttons for "Agent", "Issue Type", "Priority".

What it Measures: These are filters.

Key Insight: You can click "A103" to filter the *entire page* to show only their bad stats, or click "Critical" to see who handles the most important tickets. Use this if your boss asks a specific question during the presentation.